
RE-EXAMINING TOFU UNLEARNING: DOES LEARNING RATE ACTUALLY MATTER?

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ABSTRACT

The TOFU benchmark (*Task of Fictitious Unlearning for LLMs*) evaluates whether machine unlearning methods can make language models forget specific training information. The original paper concludes that simple baseline methods such as Gradient Ascent and Gradient Difference “fail to effectively unlearn” based on experiments using conservative learning rates. In this project, we revisit that conclusion by asking whether their pessimistic result is a property of unlearning itself or a consequence of limited hyperparameter search. We focus on Gradient Ascent unlearning on the TOFU split using Phi-1.5 and test a grid of higher learning rates than those considered in the paper. We report training losses on the forget set and evaluation losses on both forget and retain sets. Our results show that higher learning rates do not enable selective forgetting; instead, they cause catastrophic degradation on both forget and retain sets simultaneously. This confirms the TOFU paper’s core finding that simple gradient-based methods are ineffective for achieving selective unlearning, and demonstrates that hyperparameter tuning alone cannot salvage baseline approaches.

1 INTRODUCTION AND PROJECT OVERVIEW

1.1 PAPER SELECTED: TOFU: A TASK OF FICTITIOUS UNLEARNING FOR LLMs

Our project builds on *TOFU: A Task of Fictitious Unlearning for LLMs* Maini et al. (2024). TOFU defines a synthetic benchmark where large language models (LLMs) are finetuned to memorise fictitious authors and their question-answer (QA) pairs. A subset of authors forms the *forget set*, and unlearning methods are evaluated on how well they erase information about this subset while preserving performance on the remaining *retain set*.

The **core claim** we target is:

Baseline unlearning methods (e.g., Gradient Ascent) fail to effectively make language models forget information, even in a synthetic setting like TOFU.

Our question is whether this failure is due to the *algorithm* (Gradient Ascent itself), or to the *way* it was instantiated in the paper, in particular the use of conservative learning rates.

1.2 OUR GOAL: INVESTIGATING THE PESSIMISTIC CONCLUSION

In TOFU, all unlearning methods use a learning rate of 1×10^{-5} , identical to the rate used for finetuning. This is reasonable for minimisation, but may be suboptimal for Gradient Ascent, which maximises loss on the forget set. We therefore ask: is the conclusion “unlearning fails” robust to more aggressive hyperparameter choices, and can a simple baseline like Gradient Ascent be made effective by increasing the learning rate beyond the narrow range tested in the paper?

2 RELATED WORK AND CONTEXT

Machine unlearning aims to eliminate the influence of certain training data such that the resulting model behaves as if it had been trained without that data. Early work formalized this objective in simplified settings. Guo et al. (2020) prove theoretical guarantees about certified data removal in linear models, showing that exact unlearning is only possible under very restrictive assumptions. Bourtole et al. (2021) translate this goal into practice through SISA training, which approximates retraining by partitioning data and models. While effective for classification tasks, these methods don’t translate cleanly to generative language models. Recent work has focused on unlearning in LLMs.

Eldan & Russinovich (2023) showcase approximate unlearning by suppressing knowledge of “Harry Potter” in Llama-2. However, subsequent analyses show such methods often operate at the surface level and do not fully eradicate internal representations, thereby not getting rid of residual traces of the information to be forgotten.

TOFU represents a next step in this evolution by defining a controlled synthetic setting where the source of information being memorized is known and evaluating unlearning through statistical indistinguishability from a retain-only model. This ties classical unlearning definitions to modern LLM behavior.

3 HYPOTHESES AND FALSIFICATION PLAN

Alternative hypothesis The paper’s conclusion that baseline methods fail is incorrect. Their negative results come from using only conservative learning rates (10^{-5} , 10^{-6} , 5×10^{-7}). With higher learning rates (5×10^{-5} , 10^{-4} , 5×10^{-4}), Gradient Ascent can achieve effective unlearning.

Intuition and falsification strategy Gradient Ascent unlearning updates the model to increase the loss on the forget set, but with very small learning rates the updates are too small to meaningfully change what the model has memorised. To test whether stronger updates help, we sweep the learning rate over two orders of magnitude, from 10^{-5} to 5×10^{-4} , and evaluate each setting on both the forget and retain sets. A “sweet spot” where forget loss increases while retain loss stays roughly unchanged would support the alternative hypothesis. If, instead, higher learning rates always hurt both sets together, this suggests that Gradient Ascent cannot target the forget set without also damaging the rest of the model’s knowledge. Recent work also treats unlearning as a multi-task optimisation problem and introduces an adaptive learning-rate scheme to balance forgetting and retention (Bu et al., 2025), which further motivates our focus on how fixed learning-rate choices affect Gradient Ascent on TOFU.

4 METHODOLOGY

We run a small grid search over increasingly large learning rates while keeping the rest of the setup aligned with the TOFU protocol. The Gradient Ascent baseline uses $\text{LR} = 10^{-5}$ for 5 epochs on the forget set; we additionally test three higher learning rates:

Config	LR	Epochs
Paper baseline	1×10^{-5}	5
Moderate	5×10^{-5}	5
Aggressive	1×10^{-4}	5
Very aggressive	5×10^{-4}	3

We evaluate each configuration on both forget and retain sets.

Model and dataset. All experiments use Phi-1.5, following the TOFU setup. The model is trained in `bfloat16` on an NVIDIA L4 GPU. We use the official TOFU `forget01` split, where the forget set contains 40 QA pairs (2 authors) and the retain set contains 3,960 QA pairs (198 authors).

Input formatting. Each example is formatted as a Question–Answer prompt and tokenized with the `microsoft/phi-1.5` tokenizer.

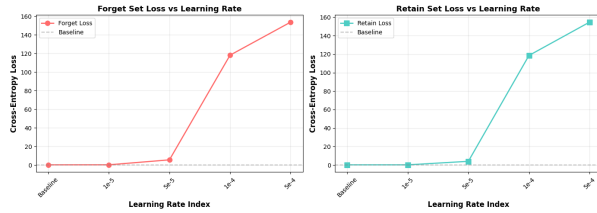
4.1 TRAINING PROTOCOL

1. **Finetuning.** Before we can “unlearn” specific authors, we first finetune Phi-1.5 on all 4,000 TOFU QA pairs ($\mathcal{S}_R \cup \mathcal{S}_F$), mirroring the TOFU protocol. We train for 5 epochs with AdamW (learning rate 2×10^{-5}), effective batch size 8 (via gradient accumulation), and standard causal language modelling loss, reaching a final training loss of 0.1636.
2. **Unlearning.** For each configuration in Table 1, we reload the finetuned checkpoint and run Gradient Ascent only on $\mathcal{S}_F$ by negating the loss on the forget set. We train for 5 epochs in all cases except the highest learning rate, where we use 3 epochs to avoid immediate collapse. Batch size, optimiser, and regularisation settings are kept fixed across runs.

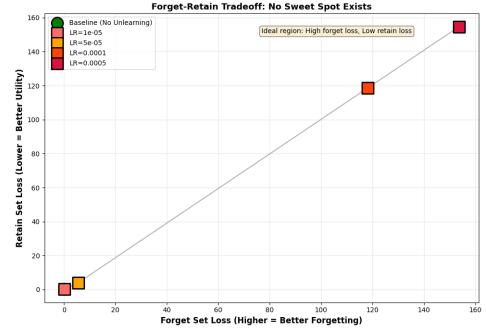
4.2 EVALUATION METRICS

We measure cross-entropy loss on both $\mathcal{S}_F$ and $\mathcal{S}_R$ before and after unlearning. For Gradient Ascent to be considered successful:

- **Forget loss** should increase substantially (the model assigns lower probability to correct answers on $\mathcal{S}_F$),



(a) Forget and Retain Loss vs Learning Rate



(b) Forget-Retain Tradeoff Space

Figure 1: (a) Both forget and retain losses increase together with higher LRs. (b) Trajectory moves diagonally rather than vertically, showing no selective forgetting regime exists.

- **Retain loss** should stay close to the finetuned baseline (the model preserves knowledge of other authors).

Loss is only a proxy for TOFU’s primary metrics (Truth Ratio and KS-test), but it is enough for our purpose: if we cannot increase loss on $\mathcal{S}_F$ without also increasing loss on $\mathcal{S}_R$, then gradient-based unlearning cannot forget $\mathcal{S}_F$ without harming $\mathcal{S}_R$. Prior work has also argued that changing model parameters alone does not guarantee desirable behaviour at the output level, especially in open-ended generative settings (Cooper et al., 2025).

5 RESULTS

5.1 BASELINE PERFORMANCE

After finetuning on the full TOFU dataset, the model achieves near-identical loss on both sets (Table 1):

- Forget loss: 0.1485, retain loss: 0.1445.

5.2 GRADIENT ASCENT UNLEARNING RESULTS

Table 1 and Figure 1a show loss values after unlearning at each learning rate.

Table 1: Loss on forget and retain sets after Gradient Ascent unlearning. Higher forget loss indicates better forgetting; lower retain loss indicates better utility preservation.

Configuration	Learning Rate	Epochs	Forget Loss	Retain Loss
Baseline (Finetuned)	—	—	0.1485	0.1445
paper_baseline	1×10^{-5}	5	0.1874 (1.3 $\times$)	0.1453 (1.0 $\times$)
moderate_5e5	5×10^{-5}	5	5.5303 (37.3 $\times$)	3.9681 (27.5 $\times$)
aggressive_1e4	1×10^{-4}	5	118.1144 (795.6 $\times$)	118.5726 (820.5 $\times$)
very_aggressive_5e4	5×10^{-4}	3	153.6653 (1035.1 $\times$)	154.6766 (1070.4 $\times$)

Key Observations: These results show that increasing the learning rate raises forget loss but causes proportional degradation in retain performance, indicating catastrophic rather than selective forgetting.

6 DISCUSSION

6.1 DID WE FALSIFY THE HYPERPARAMETER PROBLEM HYPOTHESIS?

Our alternative hypothesis was that TOFU’s pessimistic conclusion might be an artefact of conservative learning rates. Our results do not support this. We do not see any setting where higher learning rates improve unlearning on the forget set without harming the retain set. Instead, larger learning rates simply accelerate global degradation. In other words, increasing the learning rate makes Gradient Ascent more destructive, not more selective, and reinforces the TOFU finding that simple gradient-based methods damage both forget and retain knowledge simultaneously.

6.2 COMPARISON WITH THE ORIGINAL TOFU PAPER

Within the slice of the TOFU setup we reproduce, our findings are qualitatively consistent with the original paper:

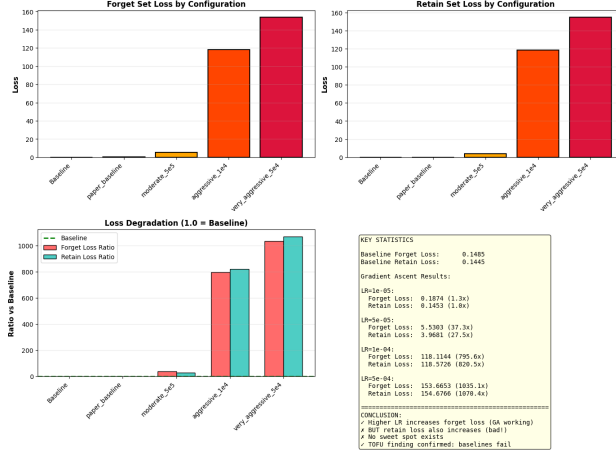


Figure 2: Summary of loss-based analysis. All configurations with higher learning rates show non-selective degradation of both forget and retain sets.

- Gradient Ascent is a blunt tool: once strong enough to noticeably damage the forget examples, it also damages retain examples.
- The model appears to lack a mechanism for *localised* forgetting of specific authors while preserving others under simple finetuning-based unlearning.

Our contribution is to make this trade-off more explicit for larger learning rates, showing that the issue is not that TOFU’s learning rate was “too small,” but that more aggressive rates are even less desirable in practice.

6.3 LIMITATIONS

- **Metrics:** We use cross-entropy loss as a proxy for TOFU’s Truth Ratio and KS-test, so we do not directly measure extractability or statistical indistinguishability.
- **Single model:** All experiments use Phi-1.5; running the same study on larger models would test how far the conclusions generalise.
- **Single method:** We only study Gradient Ascent, leaving other baselines (e.g., Gradient Difference) for future work.
- **Compute constraints:** We focus on the 1% forget split due to compute limits; behaviour at 5% and 10% may be similar or worse but we do not verify this.

7 CONCLUSION

We asked whether the TOFU paper’s conclusion that Gradient Ascent fails at machine unlearning could be explained by suboptimal learning-rate choices. Through a learning-rate sweep spanning two orders of magnitude (10^{-5} to 5×10^{-4}), we found that:

1. Higher learning rates do increase loss on the forget set, so Gradient Ascent is mechanically capable of pushing the model away from those examples.
2. Retain-set loss increases at almost the same rate, indicating catastrophic rather than selective forgetting.
3. We do not observe any “sweet spot” where unlearning is effective without destroying model utility.
4. The original paper’s conservative learning rate appears protective, keeping the model in a relatively better forget–utility trade-off region, while larger rates move it into unusable regions.

These findings support TOFU’s core message: simple gradient-based unlearning methods are fundamentally limited by entangled representations, and hyperparameter tuning alone cannot fix this. More promising directions likely require methods that explicitly disentangle representations, use auxiliary models, or optimise different objectives to achieve selective forgetting in practice.

Finally, our project highlights the value of small-scale replication and falsification. Even though our alternative hypothesis failed, the process of stress-testing it clarifies why machine unlearning remains difficult and helps sharpen future research questions.

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